

		Comparing the options, $2v \sin 15^\circ = 0.518 v$.
17	C	At top, $N + mg = mr \left(\frac{2\pi}{T} \right)^2$ $N = (77.0)(4.00) \left(\frac{2\pi}{3.70} \right)^2 - (77.0)(9.81)$ $= 133 \text{ N down}$ At bottom, $N - mg = mr \left(\frac{2\pi}{T} \right)^2$ $N = (77.0)(4.00) \left(\frac{2\pi}{3.70} \right)^2 + (77.0)(9.81)$ $= 1640 \text{ N upwards}$
18	D	$V = 5.0$